Sum and Difference Formulas

ANSWER KEY

Exact values with sum and difference formulas

1 Use a sum or difference formula to find the exact value of $\cos 75°$.

$\cos 75° = \cos(45° + 30°) = \cos 45° \cos 30° - \sin 45° \sin 30° = \frac{\sqrt{6} - \sqrt{2}}{4}$.

2 Find the exact value of $\sin \frac{\pi}{12}$.

$\sin \frac{\pi}{12} = \sin\left(\frac{\pi}{3} - \frac{\pi}{4}\right) = \sin \frac{\pi}{3} \cos \frac{\pi}{4} - \cos \frac{\pi}{3} \sin \frac{\pi}{4} = \frac{\sqrt{6} - \sqrt{2}}{4}$.

3 Find the exact value of $\tan 105°$.

$\tan 105° = \tan(60° + 45°) = \frac{\sqrt{3} + 1}{1 - \sqrt{3}}$. Rationalizing: $\frac{(\sqrt{3} + 1)^2}{(1 - \sqrt{3})(1 + \sqrt{3})} = \frac{4 + 2\sqrt{3}}{-2} = -2 - \sqrt{3}$.

Working with given values

4 Angles A and B are both in Quadrant I with $\sin A = \frac{4}{5}$ and $\cos B = \frac{5}{13}$. Find:

a $\sin(A + B)$ $\cos A = \frac{3}{5}$, $\sin B = \frac{12}{13}$, so $\sin(A + B) = \frac{4}{5} \cdot \frac{5}{13} + \frac{3}{5} \cdot \frac{12}{13} = \frac{56}{65}$.

b $\cos(A + B)$ $\cos(A + B) = \frac{3}{5} \cdot \frac{5}{13} - \frac{4}{5} \cdot \frac{12}{13} = -\frac{33}{65}$.

c the quadrant of $A + B$ Sine positive, cosine negative: Quadrant II.

5 Use a sum or difference formula to simplify:

a $\sin\left(x + \frac{\pi}{2}\right)$ $= \cos x$

b $\cos(x - \pi)$ $= -\cos x$